HOUSING
Deal Finder

COMP 381 AB1 | University of the Fraser Valley
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Motivation

- The housing market is **volatile**
- Affording a home can be a challenge
- Difficulty to tell from a listing whether it's a good price or overpaying



Our Project

- Help buyers make more informed choices
- Predict how much a home should cost based on its *features*
- Flag a home as a **"good deal"** or **"bad deal"**



OUR APPROACH

Classification

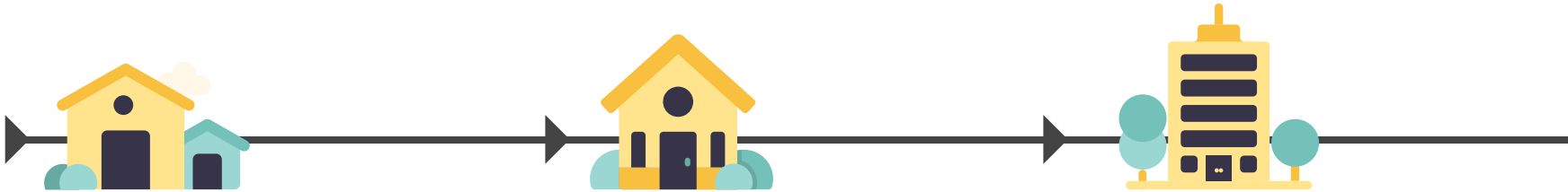
Fill in missing data

Clustering

Group data into similar areas

Linear Regression

Predict the expected price



OUR APPROACH

Identify deals

Compare asking price against predicted price



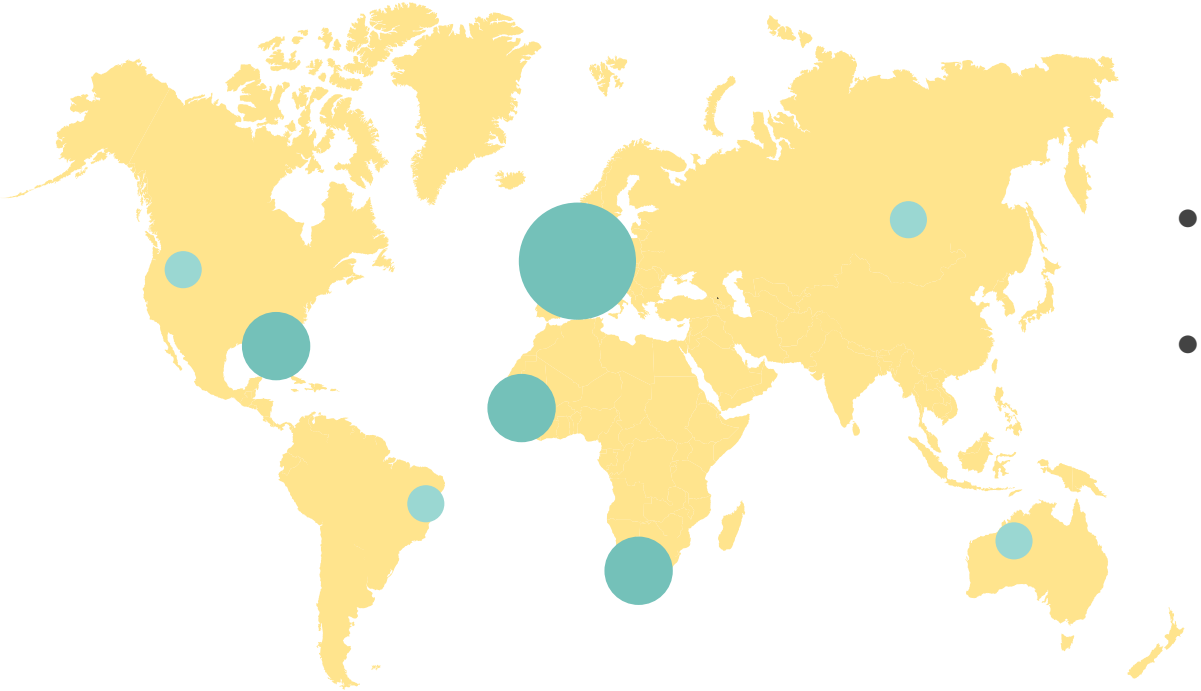
DATA PREPROCESSING

	CHANGES
BINARY	0/1
CATEGORICAL	One-hot encoded
OUTLIERS (Too big or expensive)	Removed using IQR method

FINDING THE BEST CLASSIFICATION MODEL



LOCATION-BASED CLUSTERING



- Clustered properties based on **longitude** and **latitude**
- Each cluster represents a distinct local market with its own pricing trends

PRICE PREDICTION

- Trained separate linear regression model per **cluster**
- Compared feature sets using R^2
- Combined best performing feature sets



DEAL DETECTION

GOOD DEAL

Predicted Price > Listed Price
Undervalued

BAD DEAL

Predicted Price < Listed Price
Overvalued

Core Idea: If predicted price is higher than listed price, the property is considered a good deal

Demo & Results

